DRAFT - MF algorithms for hierarchical data

Center for Veterinary Biologics - Statistics Section

November 30, 2018

Contents

INTRODUCTION **1**
Data assumptions. Nested Hierarchical structure. 1

DETERMINING MITIGATED FRACTION **3**
Rank tables. 3
Using package MF. 3
Summarizing the rank table. 4
Using package MF. 4

BOOTSTRAPPING **5**
Sampling clusters. 5
Sampling units. 5

APPENDIX **6**
Code used in this example 6
Session details for this manual. 6

INTRODUCTION

This document is intended to provide additional details regarding the algorithms used for calculating mitigated fraction values for hierarchical data as implemented in package `MF`.

A passing familiarity with the functions `MFClusHier`, `MFnest`, `MFClusBootHier`, and `MFnestBoot` is helpful in understanding the examples and underlying data assumptions in this guide. A quick start may be found at [Quick Start for Hierarchical Data].

Data assumptions. Nested Hierarchical structure.

Examples in this document expect data to be structured in a *nested hierarchical tree*, shown in Table 1. If there are only two variables (columns) or the experimental design is not nested, then the methods described here may not apply. A nested design assumes that the factor of one variable co-occur with another variable. For example, Pen D exists only within Room Z. If the experimental design of the data set is crossed (all factors of a variable co-occur with all factors of another variable), it is not appropriate to use this guide as written.

Table 1: Nested hierarchical data structure.

room	pen	litter	tx	lung
Room W	Pen A	Litter 11	vac	5.633023
				4.618143
			con	9.196908
				7.277479
		Litter 12	vac	3.760431
				3.856939
			con	6.307358
				3.521230
	Pen B	Litter 13	vac	4.356031
				6.098324
			con	7.866934
				8.339948
		Litter 14	vac	5.389661
				5.799638
			con	8.275256
				7.952039
	Pen C	Litter 15	vac	4.855498
				5.657725
			con	7.658824
				8.517004
		Litter 16	vac	4.169034
				4.837650
			con	5.822156
				7.718757
Room Z	Pen D	Litter 17	vac	3.595465
				4.921762
			con	5.222559
				5.928738
		Litter 18	vac	7.622600
				5.036120
			con	8.620428
				6.265423
	Pen E	Litter 19	vac	3.787388
				5.380694
			con	9.454640
				6.505599
		Litter 20	vac	5.300123
				4.417709
			con	7.211453
				6.351508
	Pen F	Litter 21	vac	4.690320
				6.035818
			con	6.643916
				6.995050
		Litter 22	vac	4.896763
				5.371713
			con	6.773614

DETERMINING MITIGATED FRACTION

Rank tables.

Deterimining the rank table is the first step of the algorithm. For each unique level of each variable, the following values are determined:

- `con_n` & `vac_n`: Counts of observations for each treatment (control or vaccinate) for a particular instance of unique factor level of a variable.
- `n1n2`: Product of counts, `con_n` * `vac_n`.
- `w`: Wilcoxon statistic.
- `u`: Mann-Whitney statistic.
- `con_medResp` & `vac_medResp`: Median observation for for each treatment level (control or vaccinate) in a particular instance of unique factor level of a variable.

```
## # A tibble: 12 x 10
##    room   pen   litter    con_medResp con_n     w vac_medResp vac_n  n1n2
##    <chr>  <chr> <chr>           <dbl> <dbl> <dbl>       <dbl> <dbl> <dbl>
##  1 Room W Pen A Litter 11        8.24     2     7        5.13     2     4
##  2 Room W Pen A Litter 12        4.91     2     5        3.81     2     4
##  3 Room W Pen B Litter 13        8.10     2     7        5.23     2     4
##  4 Room W Pen B Litter 14        8.11     2     7        5.59     2     4
##  5 Room W Pen C Litter 15        8.09     2     7        5.26     2     4
##  6 Room W Pen C Litter 16        6.77     2     7        4.50     2     4
##  7 Room Z Pen D Litter 17        5.58     2     7        4.26     2     4
##  8 Room Z Pen D Litter 18        7.44     2     6        6.33     2     4
##  9 Room Z Pen E Litter 19        7.98     2     7        4.58     2     4
## 10 Room Z Pen E Litter 20        6.78     2     7        4.86     2     4
## 11 Room Z Pen F Litter 21        6.82     2     7        5.36     2     4
## 12 Room Z Pen F Litter 22        6.77     1     3        5.13     2     2
## # ... with 1 more variable: u <dbl>
```

The structure of the rank table is the same regardless of the variable of interest. That is, whether a user is interested in the value of mitigated fraction for variable "pen" or "litter" does not affect the content of rows or columns, assuming the same input data.

Using package MF.

Rank tables can be output with the following usage in package `MF`:

- `MFClusHier(...)$MFh`
- `MFh(...)`
- `MFClusBootHier(...)$MFhBoot`
- `MFhBoot(...)`

Summarizing the rank table.

With a detailed rank table, it is possible to calculate the mitigated fraction for a variety of variable choices. This is the second step of the algorithm. For each variable of interest the following summations are calculated for each factor of that variable.

- `N1N2`: Sum of the `n1n2` values from the rank table for the factor level of that row.
- `U`: Sum of the `u` values from the rank table for the factor level of that row.
- `con_N` & `vac_N`: Sum of the counts from the rank table matching the particular factor level of the row.
- `con_medResp` & `vac_medResp`: Median of responses for each comparison group matching the particular factor level of the row.
- `MF`: $2 * U/N1N2$.

The following example looks at the mitigated fraction of factors in variable "pen". We can see that the values for "N1N2" have been summed from the previous rank table for all cases of "Pen A", "Pen B", etc.

```
## # A tibble: 6 x 9
##   variable level    MF  N1N2     U con_N vac_N con_medResp vac_medResp
##   <fct>    <chr> <dbl> <dbl> <dbl> <dbl> <dbl>       <dbl>       <dbl>
## 1 pen      Pen A  0.5      8     6     4     4        6.79        4.24
## 2 pen      Pen B  1        8     8     4     4        8.11        5.59
## 3 pen      Pen C  1        8     8     4     4        7.69        4.85
## 4 pen      Pen D  0.75     8     7     4     4        6.10        4.98
## 5 pen      Pen E  1        8     8     4     4        6.86        4.86
## 6 pen      Pen F  1        6     6     3     4        6.77        5.13
```

It is possible to perform this summary without regard for variables to get an overall mitigated fraction value.

```
## # A tibble: 1 x 9
##   variable level    MF  N1N2     U con_N vac_N con_medResp vac_medResp
##   <fct>    <chr> <dbl> <dbl> <dbl> <dbl> <dbl>       <dbl>       <dbl>
## 1 All      All   0.870    46    43    23    24        7.21        4.91
```

Using package MF.

- `MFClusHier(...)`: calculates rank table and MF table as a single step.
- `MFnest(...)`: calculates MF table from rank table.
- `MFClusBootHier(...)`: calculates rank table and MF table as a single step.
- `MFnestBoot(...)`: calculates MF table from rank table.

BOOTSTRAPPING

Bootstrapping can occur in one of two points of consideration: at *clusters* or at *units*.

Sampling clusters.

A *cluster* refers to the full grouping designation of each observation. For example, in Table 1 the cluster of the first observation is Room W/Pen A/Litter 11, and there are four observations within that cluster. Bootstrapping clusters samples from the unique clusters with replacement, so every instance will have the same number of clusters as the original data, but they may not be unique. The example below shows one possiblility of a bootstrap instance from the example data.

```
## # A tibble: 12 x 4
##    bootID room   pen   litter
##     <int> <chr>  <chr> <chr>
##  1      1 Room W Pen A Litter 11
##  2      1 Room W Pen B Litter 13
##  3      1 Room W Pen B Litter 14
##  4      1 Room W Pen C Litter 15
##  5      1 Room W Pen C Litter 15
##  6      1 Room W Pen C Litter 16
##  7      1 Room Z Pen E Litter 20
##  8      1 Room Z Pen E Litter 20
##  9      1 Room Z Pen E Litter 20
## 10      1 Room Z Pen F Litter 21
## 11      1 Room Z Pen F Litter 22
## 12      1 Room Z Pen F Litter 22
```

During bootstrapping of clusters, the observations within that cluster do not change composition. That is, Room W/Pen A/Litter 11 continues to have two vaccinates (5.63, 4.62) and two controls (9.20, 7.28).

Sampling units.

Bootstrapping at the unit level involves sampling with replacement the observed values within a particular treatment group. The mix of control and vaccinate groups remains constant. This affects the final calculation of MF because the `U` value may be different.

APPENDIX

Code used in this example

```
a <- data.frame(
 room = paste('Room', rep(c('W', 'Z'), each = 24)),
 pen = paste('Pen', rep(LETTERS[1:6], each = 8)),
 litter = paste('Litter', rep(11:22, each = 4)),
 tx = rep(rep(c('vac', 'con'), each = 2), 12),
 stringsAsFactors = FALSE
 )
set.seed(76153)
a$lung[a$tx == 'vac'] <- rnorm(24, 5, 1.3)
a$lung[a$tx == 'con'] <- rnorm(24, 7, 1.3)
a <- a[-48,]

# DETERMINING MITIGATED FRACTION
thismfh <- MFh(formula = lung ~ tx + room/pen/litter, data = a)
MFnest(thismfh, which.factor = "pen")
MFnest(thismfh, which.factor = "All")

# BOOTSTRAPPING
MFClusBootHier(formula = lung ~ tx + room/pen/litter,
                                  data = a, boot.unit = FALSE,
                                  boot.cluster = TRUE,  alpha = 0.1,
                                  which.factor = c('room', 'litter', 'All'))
mfboot_multiple$MFhBoot$bootmfh %>%
  filter(bootID == 1) %>%
  select(c('bootID', 'room', 'pen', 'litter'))
```

Session details for this manual.

```
sessionInfo()
```

```
## R version 3.4.4 (2018-03-15)
## Platform: x86_64-w64-mingw32/x64 (64-bit)
## Running under: Windows 10 x64 (build 14393)
##
## Matrix products: default
##
## locale:
## [1] LC_COLLATE=English_United States.1252
```

```
## [2] LC_CTYPE=English_United States.1252
## [3] LC_MONETARY=English_United States.1252
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.1252
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] bindrcpp_0.2.2   MF_4.3.5         forcats_0.3.0    stringr_1.3.1
##  [5] dplyr_0.7.8      purrr_0.2.5      readr_1.1.1      tidyr_0.8.2
##  [9] tibble_1.4.2     ggplot2_3.0.0    tidyverse_1.2.1  kableExtra_0.9.0
##
## loaded via a namespace (and not attached):
##  [1] Rcpp_1.0.0        cellranger_1.1.0  pillar_1.2.3
##  [4] compiler_3.4.4    plyr_1.8.4        bindr_0.1.1
##  [7] tools_3.4.4       digest_0.6.15     lubridate_1.7.4
## [10] jsonlite_1.5      gtable_0.2.0      evaluate_0.10.1
## [13] nlme_3.1-131.1    viridisLite_0.3.0 lattice_0.20-35
## [16] pkgconfig_2.0.1   rlang_0.3.0.1     cli_1.0.0
## [19] rstudioapi_0.7    yaml_2.1.19       haven_1.1.2
## [22] withr_2.1.2       httr_1.3.1        xml2_1.2.0
## [25] knitr_1.20        hms_0.4.2         grid_3.4.4
## [28] rprojroot_1.3-2   tidyselect_0.2.5  glue_1.3.0
## [31] R6_2.2.2          readxl_1.1.0      rmarkdown_1.10
## [34] modelr_0.1.2      magrittr_1.5      backports_1.1.2
## [37] scales_0.5.0      htmltools_0.3.6   assertthat_0.2.0
## [40] rvest_0.3.2       colorspace_1.3-2  utf8_1.1.4
## [43] stringi_1.1.7     lazyeval_0.2.1    munsell_0.5.0
## [46] broom_0.5.0       crayon_1.3.4
```